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P1.

(a)

Price	$\pi_1=0.3$, i=1	$\pi_2=0.4$, i=2	$\pi_3=0.3$, i=3
8	5	10	15
1	1.1	1.1	1.1
1	0	0	3

The market is complete

for state-3 : $q_3=\frac{1}{3}$

for state-2 : $q_2=\frac{5}{5}-\frac{1}{1.1}-\frac{2}{3}=\frac{4}{165}$

for state-1 : $q_1=\frac{1}{1.1}-\frac{1}{3}-\frac{4}{165}=\frac{91}{165}$

(b)

price	i=1	i=2	i=3
P	3	0	0

$P=3q_1=\frac{91}{55}$

(c) $q^i=\frac{1}{1+r_f}\pi_i^{RN}$ $\pi_1^{RN}=(1+r_f)q_1=1.1\cdot\frac{91}{165}=\frac{91}{150}$ $\pi_2^{RN}=(1+r_f)q_2=\frac{2}{75}$

$\pi_3^{RN}=(1+r_f)q_3=\frac{11}{30}$

P2 (a) $P_0=100P_1+100P_2+1100P_3=940$

(b) $P_0=100P_2+100P_3=150$

(c) $P_0=100P_2-100P_3=10$

P3 $\because E(mR_f)=1$ $\therefore R_f\cdot E(m)=1$ $\therefore E(m)=\frac{1}{R_f}$

$\because E(mR_m)=1$ $\therefore E(m)E(R_m)+cov(m,R_m)=1$

$\therefore \frac{E(R_m)}{R_f}+cov(m,R_m)=1$

$\therefore E(R_m)=R_f-cov(m,R_m)R_f$

$\because E(mR)=1$ $\therefore E(R)=R_f-cov(m,R)R_f$

let $\beta=\frac{cov(m,R)}{cov(m,R_m)}$ $\therefore \beta=\frac{R_f-E(R)}{R_f-E(R_m)}$

$E(R)=\beta(E(R_m)-R_f)+R_f$

$=R_f+\beta(E(\tilde{R}_m)-r_f)$

$\beta=\frac{cov(m,R)}{cov(m,R_m)}=\frac{cov(R,R_m)}{var(R_m)}$